A Bilinear Isochronous Oscillator: Global Action–Angle Coordinates and Quantum Interface Spectrum

Source-local theorem reconstruction

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Abstract

We close the classical and quantum dynamics of a joined pair of quadratic half-wells. Every nonzero classical orbit has one energy-independent least period exactly when both stiffnesses are positive. Its global action–angle chart is C^1 and piecewise analytic, but detects unequal stiffness at second order. The natural Friedrichs quantization has simple discrete spectrum characterized completely by a parabolic-cylinder interface Wronskian. Flat-side, free, zero-energy, and equal-frequency boundaries are kept explicit.

1 The common-period owner

For $x_+ = \max\{x, 0\}$, $x_- = \min\{x, 0\}$, and $\omega_+, \omega_- \geq 0$, consider

$$H(x,p) = \frac{p^2}{2} + V(x), \qquad V(x) = \frac{\omega_+^2 x_+^2 + \omega_-^2 x_-^2}{2}. \tag{1}$$

The force is continuous and globally Lipschitz, hence the flow is unique and complete, and H is conserved.

Theorem 1 (all-energy isochrony)**.** *Every nonzero orbit of (1) is bounded and periodic with a common least period if and only if $\omega_+, \omega_- > 0$. In that chamber,*

$$T = \pi\left(\frac{1}{\omega_+} + \frac{1}{\omega_-}\right), \qquad J(E) = \frac{E}{2}\left(\frac{1}{\omega_+} + \frac{1}{\omega_-}\right), \qquad \Omega = \frac{dE}{dJ} = \frac{2}{1/\omega_+ + 1/\omega_-}. \tag{2}$$

Proof. At $E > 0$, write $B = \sqrt{2E}$ and start at $(0, B)$. The right half-excursion is

$$x = \frac{B}{\omega_+}\sin(\omega_+ t), \quad p = B\cos(\omega_+ t), \quad 0 \leq t \leq \frac{\pi}{\omega_+}.$$

Starting next from $(0, -B)$, with $s = t - \pi/\omega_+$, the left one is

$$x = -\frac{B}{\omega_-}\sin(\omega_- s), \quad p = -B\cos(\omega_- s), \quad 0 \leq s \leq \frac{\pi}{\omega_-}.$$

They give the least period in (2). Their two half-ellipse areas are $\pi E/\omega_+$ and $\pi E/\omega_-$, proving the action formula. If a frequency vanishes, an orbit entering that flat half-axis with nonzero speed escapes linearly; if both vanish the motion is free. This proves the necessity. □

2 Seam-compatible action–angle coordinates

Put $\theta_+ = \pi\Omega/\omega_+$, $B = \sqrt{2\Omega J}$, and measure $\theta = \Omega t$ from $(0, p > 0)$. The inverse chart is

$$\begin{aligned} 0 \leq \theta \leq \theta_+ : &\quad x = \frac{B}{\omega_+}\sin\frac{\omega_+\theta}{\Omega}, \quad p = B\cos\frac{\omega_+\theta}{\Omega}, \\ \theta_+ \leq \theta \leq 2\pi : &\quad x = -\frac{B}{\omega_-}\sin\frac{\omega_-(\theta - \theta_+)}{\Omega}, \quad p = -B\cos\frac{\omega_-(\theta-\theta_+)}{\Omega}. \end{aligned} \tag{3}$$

Values and first J, θ derivatives agree at the two seams. On each open piece,

$$x_\theta p_J - x_J p_\theta = 1,$$

so (3) is a global C^1, piecewise-analytic symplectic chart from $(0,\infty) \times \mathbb{R}/(2\pi\mathbb{Z})$ to the punctured plane, with $dx \wedge dp = d\theta \wedge dJ$ and $\dot{\theta} = \Omega$. At θ_+, the one-sided values of $p_{\theta\theta}$ are $B(\omega_+/\Omega)^2$ and $B(\omega_-/\Omega)^2$; the chart is not C^2 unless the frequencies agree. Each normalized seam-to-seam half-flow matrix is $-I_2$. Thus the common-period map and its derivative are exactly I_2. This is the seam-compatible action-angle owner.